

# Infrared Small Target Detection Based on Holistic Interframe Interaction and Spatiotemporal Local Contrast Method

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**Abstract**—Infrared small target detection plays a crucial role in infrared search and tracking systems. However, current detection methods are limited by the small target size and low signal-to-noise ratio of infrared imagery. Furthermore, motion features for target detection are difficult to extract using simple frame subtraction due to poor imaging conditions. Therefore, we focus on the holistic interframe interaction to enhance the temporal feature and propose a spatiotemporal local contrast method in this letter. First, the motion-enhanced density peak clustering is employed to determine the robust localization of candidate targets, in which the density feature maps are generated by the preprocessing of non-consecutive three frames difference after image registration. Second, to reliably exploit interframe interactions across both non-consecutive and successive frames, a temporal domain saliency map is computed based on local regions from successive frames. Moreover, a spatial domain saliency map is obtained using a novel tri-layer local contrast measure. By fusing results from both domains, the infrared small targets are detected through adaptive threshold segmentation. Experimental results on four real sequences demonstrate that the proposed method can achieve better detection performance by target enhancement and background suppression than other spatiotemporal algorithms.

**Index Terms**—Infrared small target detection, density peak clustering, interframe interaction, spatiotemporal local contrast.

## I. INTRODUCTION

INFRARED (IR) imaging system has the advantages of strong anti-interference ability, strong concealment, and all-weather work [1]. Therefore, it is widely used in various fields of defense and civilian applications [2], such as forest fires detection, early warning and reconnaissance tasks. The moving IR targets, such as small aircraft and unmanned aerial vehicles, are usually observed at long imaging distance, resulting in the lack of structure and texture information [3], and appearing as point-like targets with low signal-to-clutter ratio (SCR), which are easily buried in background with interference. Therefore, the IR small moving objects detection in complex background conditions remains a challenging task.

The existing infrared small target detection (ISTD) methods mainly include single frame and sequential detection methods. Sequential methods incorporate temporal cues across consecutive frames [4], enabling the extraction of more comprehensive information about both targets and background, so it serves as the primary focus of this study.

Inspired by the characteristics of the human visual system

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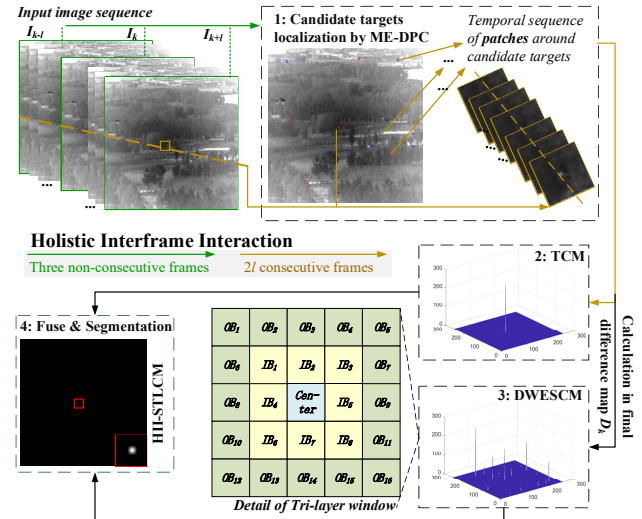


Fig. 1. Flowchart of the proposed HII-STLCM. The green and yellow line denotes the interframe interaction with non-consecutive and consecutive frames, respectively.

(HVS), the single frame local contrast methods have gained widespread attention and progress. Chen et al. [5] proposed a local contrast measure (LCM), an eight-direction and double-window filter was first constructed to enhance the target. Wu et al. [6] designed a computational LCM structure by double neighborhood gradient method (DNGM). Additionally, Xia et al [7] proposed a new dissimilarity degree called the local energy factor (LEF) to measure the brightness difference between small targets and backgrounds. Han et al [8] firstly proposed the tri-layer local contrast measure (TLLCM), which performs Gaussian filtering on the center layer of the window to remove noise. Furthermore, Kou et al [9] introduced density peaks clustering (DPC) in the tri-layer method. However, the use of fixed rectangular windows in the abovementioned algorithms limits their ability to accurately measure contrast between the target and background, leading to reduced saliency for true targets with complex or varying pixel distributions. Furthermore, these methods extract features solely on spatial domain, in which the low SCR conditions limit their detection performance.

Compared to the single frame methods, the sequential spatiotemporal local contrast methods which use both spatial and temporal information of multiple frames are now widely adopted. Deng et al. first proposed a spatial-temporal local contrast filter (STLCF) [10], which integrates temporal motion features of the target with spatial contrast features to detect moving IR targets. Additionally, Du et al proposed a spatiotemporal local difference measure (STLDM) [11]. Zhang et al proposed a spatial and temporal vector difference measure (STVDM) [12] for IR moving object detection. Similarly, Yu et al proposed a strengthened spatiotemporal tri-layer local contrast method

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[13]. Furthermore, Ma et al designed a coarse-to-fine structure of moving targets detection (MCFS) [1], combining the temporal Kurtosis features and spatial saliency map by the multiscale tri-layer local contrast features.

As mentioned above, so many methods were developed to deal with specific application scenarios of ISTD, and generally some of them can achieve ideal results, but there are still issues need to be addressed. In terms of detecting moving targets, the shape of target may change rapidly between consecutive frames, and abnormal conditions such as camera shake may also occur. In addition, existing methods often extract temporal information only from non-consecutive or successive frames, which may fail to capture richer interframe temporal correlations. Moreover, the spatial information utilized in these methods is relatively coarse, especially for point-like IR targets, which do not fit well into the predefined local contrast windows, resulting in higher false alarm rate in complex backgrounds.

Therefore, in view of the abovementioned shortcomings, we propose an ISTD method based on holistic interframe interaction and spatiotemporal local contrast method (HII-STLCM), aiming to fully exploit the temporal information. The main contributions of the letter are summarized as follows:

- 1) The motion-enhanced density peak clustering (ME-DPC) is designed for fast candidate target extraction and location using a non-consecutive three-frame difference, and the further temporal contrast map (TCM) is extracted on successive frames progressively, thus strengthening holistic temporal information extraction across interframe.
- 2) Obtaining the double weight enhanced spatial contrast map (DWESCM) by improved tri-layer local contrast measure, which effectively suppress background clutter and enhance the saliency of targets. The spatiotemporal saliency map is enhanced by jointly incorporating TCM and DWESCM.
- 3) Extensive experiments demonstrate that our method outperforms existing approaches to accurately detect targets.

## II. PROPOSED METHOD

The proposed HII-STLCM includes four steps: 1) Candidate targets are located by using the ME-DPC operation on non-consecutive frames; 2) TCM are extracted by the local gray intensity on successive frames 3) DWESCM is computed on the  $k$ -th difference map. 4) The fusion saliency map of targets is obtained by integrating the TCM with DWESCM, and the targets are decided using a threshold segmentation. The flowchart of the proposed method is shown in Fig. 1.

### A. Candidate Targets Localization by ME-DPC

1) *Image Preprocessing*: The original images containing IR small targets are not well-suited as input density maps for DPC due to their low SNR and background clutter. As shown in the bi-temporal IR images in Fig. 2, the shake of camera and other disturbances cause both target and background pixels to move; however, their motion directions often differ. The inconsistency in imaging coordinate systems can adversely affect detection performance. Hence, image registration is required for the sequence prior to performing detection [14].

As illustrated in step1 in Fig. 2, firstly, we select two non-consecutive frames, one frame before and another frame after the current frame with an interval  $l$ , and a 2D rigid transformation is then estimated for camera motion compensation

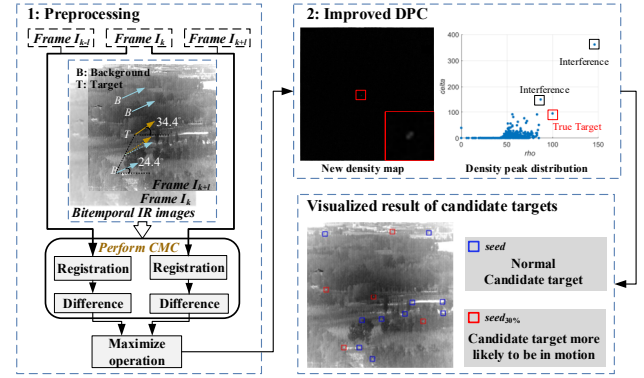


Fig. 2. The flow chart of the candidate targets localization by ME-DPC.

(CMC). The transformation matrix is calculated as [14]:

$$I'_r(x', y') = I_r(x, y), \quad r = k \pm l \quad (1)$$

where the transformed image  $I'$  is related to the original image  $I$  through the transformation of pixel coordinates.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (2)$$

where  $\theta$  represents the rotation angle,  $t_x$  and  $t_y$  define the translations along the horizontal and vertical axes, respectively.

Calculating interframe differences offers an effective interaction of emphasizing targets with motion patterns distinct from the background. However, using only a two-frame difference may lead to poor detection of slow-moving targets, since their small displacements often result in overlapping regions, which is canceled out during the subtraction process [14]. To address this, a three-frame difference strategy is adopted for background suppression.

Firstly, the absolute differences between the two registered images and current frame are computed. Then we combine them by using a pixel-wise maximum operation to preserve more potential candidate targets at this initial localization stage. This can be mathematically described as:

$$D_{k+l} = |I'_{k+l} - I_k|, \quad D_{k-l} = |I_k - I'_{k-l}| \quad (3)$$

$$D_k(x, y) = \max(D_{k+l}(x, y), D_{k-l}(x, y)) \quad (4)$$

where  $I_k$ ,  $I'_{k+l}$  and  $I'_{k-l}$  denote the current frame with index  $k$ , the previous registered frame, and the next registered frame with interval  $l$ ,  $D_{k+l}$  and  $D_{k-l}$  represent their absolute differences maps, respectively.  $D_k$  is the resulting final difference map.

2) *Identification of Candidate targets by improved DPC*: Inspired by the DPC method [8], the density  $\rho_i$  and  $\delta$ -distance of the pixel in the image are defined as:

$$\rho_i = D_i, \quad \delta_i = \min_{j: p_j > p_i} (d_{ij}) \quad (5)$$

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (6)$$

where  $D_i$  is the gray value of pixel  $i$  of the difference map, which is derived from the intersection of the preprocessing results of two image pairs.  $\delta_i$  is the minimum distance of  $d_{ij}$  between pixel  $i$  and any pixel with a higher density,  $(x_i, y_i)$  and  $(x_j, y_j)$  denote the position of pixel  $i$  and  $j$ , respectively. The density peak  $p$  is calculated as follows:

$$p = \rho_{n,i}^2 * \delta_{n,i} \quad (7)$$

$$\rho_{n,i} = \frac{\rho_i}{\max(\rho_i(k))}, \quad \delta_{n,i} = \frac{\delta_i}{\max(\delta_i(k))} \quad (8)$$

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TABLE I  
COMPARISON WITH CONSIDERED METHODS ON THE FOUR SEQUENCES. THE BEST RESULTS ARE MARKED IN **BOLDFACE**

| Metrics | Sequence | NWTH          | LEF    | TLLCM  | STLDM  | STLCF  | STRL-LBCM | MCFS          | HII-STLCM     |
|---------|----------|---------------|--------|--------|--------|--------|-----------|---------------|---------------|
| SCRG    | Seq1     | 47.406        | 450.99 | 134.89 | 129.39 | 330.54 | 50.438    | 221.17        | <b>620.27</b> |
|         | Seq2     | 26.218        | 32.564 | 4.3141 | 4.5086 | 22.805 | 2.1443    | 21.638        | <b>34.477</b> |
|         | Seq3     | 15.536        | 17.653 | 14.010 | 14.926 | 13.619 | 11.708    | 8.9879        | <b>18.631</b> |
|         | Seq4     | 132.12        | 177.67 | 1.3888 | 191.45 | 128.84 | 76.600    | 18.123        | <b>193.87</b> |
| BSF     | Seq1     | 0.3888        | 0.3913 | 0.3666 | 0.3955 | 0.4179 | 0.9595    | <b>2.7994</b> | <b>1.1229</b> |
|         | Seq2     | 0.7378        | 4.1311 | 0.7207 | 0.7635 | 0.6802 | 1.2992    | 0.7323        | <b>4.2960</b> |
|         | Seq3     | 5.3525        | 4.7658 | 0.6966 | 2.2373 | 0.5742 | 5.1885    | 4.7163        | <b>7.1684</b> |
|         | Seq4     | 13.552        | 30.613 | 2.0137 | 24.659 | 3.4074 | 15.567    | 7.5860        | <b>48.086</b> |
| Time/s  | Average  | <b>0.0054</b> | 4.1973 | 0.6019 | 1.1099 | 0.2179 | 1.1597    | 0.3509        | 0.8272        |

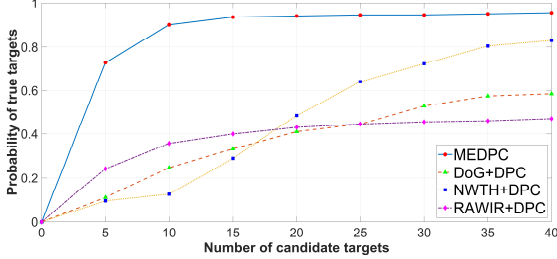


Fig. 3. Probability that candidate targets contain the true target.

where the  $\rho_{n,i}$  and  $\delta_{n,i}$  denote the normalized density and distance, by which the noise is smoothed and suppressed.

Since the gray intensity in difference map reflects the likelihood of motion, it is squared to further emphasize motion regions and moving target visibility. Finally, as shown in step2 in Fig. 2, all density peaks are sorted in descending order. The top  $n$  highest-density peaks are selected as candidate targets:

$$seed = \{(x, y)_1, \dots, (x, y)_m, \dots, (x, y)_n\} \quad (9)$$

where  $seed$  is the set of the top  $n$  candidate target positions.

### B. Temporal Contrast Map

The ME-DPC can get the initial candidate targets which have a higher chance in motion. However, as shown in Fig.2, some interferences may have similar density peaks as true target, leading to incorrect target identification, so that we design the TCM to further extract the temporal feature of moving IR target.

When detecting moving targets with too slow or fast speed in adjacent frames, the targets may be suppressed or incorrectly re-detected across frames. Therefore, we compute the TCM over successive frames in the neighborhood of each candidate target [12]. The TCM is defined as follows:

$$TCM_k = (R_{\max} - R_{\min})^2 \quad (10)$$

$$R_{\max} = \max\{m_{k-l}, \dots, m_k, \dots, m_{k+l}\} \quad (11)$$

$$R_{\min} = \min\{m_{k-l}, \dots, m_k, \dots, m_{k+l}\} \quad (12)$$

$$m_n = \max(P_{f_n}(x_t, y_t)), \quad \forall (x_t, y_t) \in seed \quad (13)$$

where  $P_{f_n}$  is the region of central patch of frame  $n$  at position  $(x_t, y_t)$ , the size of the patch is  $5 \times 5$  pixels, and  $\{m_{k-l}, \dots, m_k, \dots, m_{k+l}\}$  is the temporal patch sequence, the central index  $k$  is same as the index of frame  $I_k$  in Eq. (3).  $R_{\max}$  and  $R_{\min}$  are the largest and smallest values in the sequence.

### C. Double Weight Enhanced Spatial Contrast Map

Inspired by the DNGM [7], a three-layer filtering template is proposed for improved spatial local contrast extraction, which further filter out false targets from initial candidate targets. The detail of this tri-layer window is illustrated in Fig. 1, the entire window is divided into three regions, where the center cell is

used to capture the main energy of the target. The inner layer contains 8 inner blocks (IB), and the outermost layer contains the 16 outer blocks (OB), respectively. The size of each cell is  $3 \times 3$  pixels.

1) *Center layer*: The IR small target approximately follows an irregular 2D Gaussian distribution [9]. Therefore, we use the Gaussian-filter for target enhancement in the center layer. The filter is defined as:

$$I_c^{(k)} = \sum_{m=-1}^1 \sum_{n=-1}^1 C_G(m, n) * D_k(x_t + m, y_t + n), \quad ((x_t, y_t) \in seed) \quad (14)$$

where  $(x, y)$  are the coordinates of the center layer.  $C_G$  is the Gaussian filter.  $I_c^{(k)}$  is the filtered map of difference map  $D_k$ .

2) *Inner and Outermost layer*: To enhance the target contrast and suppress noise effectively, for the inner layer, we calculate the difference between the center cell  $T$  and inner cell, and the weight of inner  $W_{IB}$  is obtained as:

$$W_{IB} = \begin{cases} m_{I_c^{(k)}} - \max(m_{IB_i}), & m_{I_c^{(k)}} > \max(m_{IB_i}) \\ 0, & \text{else} \end{cases} \quad (15)$$

where  $m_{I_c^{(k)}}$  and  $m_{IB_i}$  are the mean value of the central cell filter by Gaussian and  $i$ -th inner layer cell of difference map  $D_k$ .

The DNGM method assumes that the central cell containing the target has the highest mean gray-level intensity compared to the outermost layer. However, infrared point-like targets, typically occupying  $3 \times 3$  pixels region or smaller, may appear dimmer than the surrounding background, which usually spans a larger area and exhibits higher average brightness. This contrast may result in false detections. Generally, a smaller mean gray-level difference between the center and the outermost layers indicates higher similarity [9]. Therefore, we replace the maximum value in DNGM with the mean gray-level difference, providing a more flexible criterion for determining whether a cell contains a target. For the outermost layer, the weight  $W_{OB}$  is defined as:

$$W_{OB} = \frac{1}{8} \sum_j \left| \frac{m_{I_c^{(k)}} - m_{OB_j}}{m_{I_c^{(k)}}} \right| \quad (16)$$

where  $m_{OB_j}$  is the mean value of the  $j$ th outermost cell. And the final DWESCM of frame index  $k$  can be defined as:

$$DWESCM_k = W_{IB} * W_{OB} \quad (17)$$

### D. Foreground Object Segmentation

The final HII-STLCM is derived by fusing DWESCM and TCM through a multiplication-based strategy, defined as:

$$HII-STLCM = \beta \cdot DWESCM_k(ct) \times TCM_k(ct), \quad \forall ct \in seed \quad (18)$$

where  $ct$  define the candidate targets. The weight  $\beta$  is assigned



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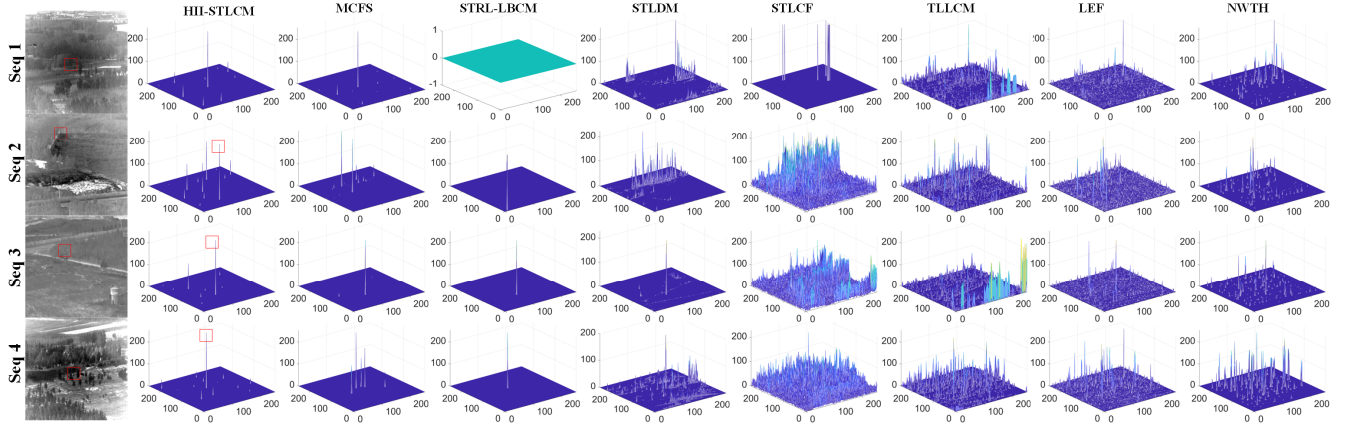


Fig. 4. Representative detection comparative results of for scenarios in Seq 1-4.

to reasonably enhance potential moving IR target regions, which can be defined as:

$$\beta = \begin{cases} \beta_1, & ct \in seed_{30\%} \\ \beta_2, & ct \notin seed_{30\%} \end{cases} \quad (19)$$

$$seed_{30\%} = \{seed_q | q > 0.7n\} \quad (20)$$

where  $seed_{30\%}$  refers to coordinates of the top 30% density in  $seed$ , the members in which are more likely to be the true moving targets. Then, considering that the saliency map has different levels of background clutter, the IR targets are segmented by an adaptive thresholding method [8].

### III. EXPERIMENTS AND ANALYSIS

To validate the effectiveness of the proposed method, we conducted experiments on four publicly available real IR sequences [16]. Each sequence contains 300 frames with air-to-ground backgrounds and potential targets. All images have a resolution of  $256 \times 256$  pixels, and the targets are not larger than  $3 \times 3$  pixels. The experiments were performed on a PC equipped with a 3.7 GHz Intel i9 CPU and 32 GB of RAM.

The intervals  $l$  is set to 5, and the weight  $\beta_1$  and  $\beta_2$  are set to 4 and 1, respectively, to yield optimal performance based on the experimental results.

#### A. Evaluation Metrics and Comparison Methods

As the commonly used metrics for evaluating ISTD performance, the signal-to-clutter ratio gain (SCRG) is used to evaluate the target enhancement ability, and background suppression factor (BSF) is used to evaluate the background suppression performance [17]. We also use receiver operating characteristic (ROC) curve and area under curve (AUC) to quantitatively assess the overall detection performance.

In this letter, single-frame detection algorithms (NWTB [18], TLLCM [8], and LEF [7]), and the spatial-temporal joint methods (STLCF [10], STLDM [11], STRL-LBCM [19], and MCFS [1]) are chosen as the comparison methods.

#### B. Experimental Results and Analysis

1) *Performance of the ME-DPC*: To evaluate the effectiveness of identifying candidate targets, we conduct experiment using our ME-DPC to quantitatively assess its ability to extract the true targets. We also test original DPC on raw IR images, as well as its variants with DoG filtering [19] and NWTB filtering [9], the experimental results are shown in Fig. 3. It can be seen

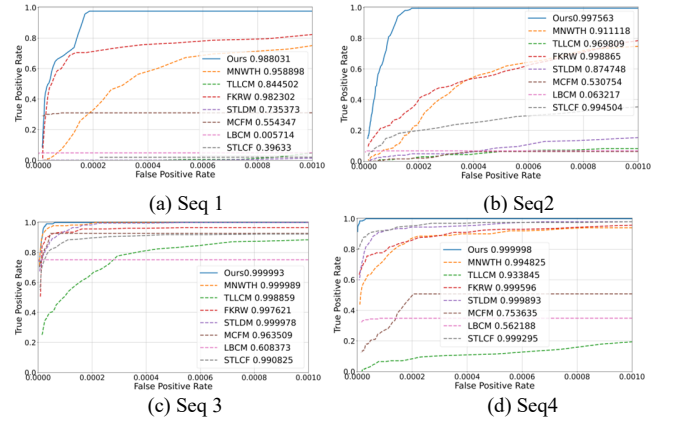


Fig. 5. ROC curves of different methods for the Seq 1-4.

that the probability that the abovementioned four approaches can capture targets increases logarithmically with the number of candidate targets. Among the four methods, original DPC has the worst performance. In contrast, our ME-DPC, which integrates registration, non-consecutive three-frame difference and DPC, demonstrates the best effect. When the candidate target  $n$  is set to 15, 20, and 25, the probabilities that ME-DPC can capture correct targets are 93.5%, 94.1%, and 94.5%, respectively. Furthermore, it can be seen that under the same candidate number settings  $n$ , the proposed HII-STLCM has an even higher probability of capturing the true candidate targets.

2) *Vision comparison*: Representative detection results for all sequences are shown in Fig. 4. As can be observed, TLLCM fails to effectively detect true targets from complicated background, this limitation arises from its fixed tri-layer windows, which are not well-suited to the point-like small targets. In contrast, HII-STLCM can perform well in suppressing clutter and highlighting targets. Even in the Seq.1 and Seq.2, which feature highly complex backgrounds, our method enhances target saliency significantly, despite interference from background structures with similar characteristics. Other methods, by comparison, largely fail to detect the correct targets in these cases. Besides, the proposed HII-STLCM and some other methods like MCFS and STRL-LBCM all manage to correctly detect targets in Seq.3 and Seq.4, but our results yield highly salient outputs that separate true targets from background clutter.

3) *Quantitative Evaluation*: The quantitative results of SCRG, BSF and running time are presented in Table I. It can be clearly seen that the proposed method has achieved notable ad-

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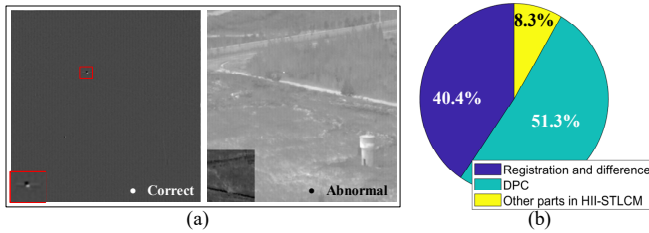


Fig. 6. Test results of the discussion part. (a) Compare of the correct and abnormal frame difference. (b) The proportion of processing time of component parts in proposed HII-STLCM.

vantages in terms of SCRG, demonstrating superior target enhancement capability due to its holistic interframe interaction architecture. LEF and STLDM exhibit the similar performance in Seq.3 and Seq.4, however, their results in Seq.1 and Seq.2 are relatively poor. Regarding BSF, although our HII-STLCM does not achieve the maximum across all sequences, it also reaches very competitive results. Furthermore, in terms of average detection time across four sequences, compared to similar algorithms, HII-STLCM is faster than STLDM but slightly slower than MCFS. Nevertheless, its overall performance significantly outperforms that of MCFS.

The ROC curves of different methods are shown in Fig. 5, with the corresponding AUC values listed in the legend, it is evident that the ROC curve of the proposed method consistently lies in the upper left corner across all sequences, indicating superior detection accuracy. These results demonstrate that our method balances robustness and computational efficiency.

4) *Performance of the camera motion compensation (CMC)*: Our method incorporates a candidate selection mechanism within a dynamic spatiotemporal ISTD method, it is evident that the effectiveness of which largely depends on the registration accuracy used for CMC. We try the ablation studies on different registration methods on Seq.1, including MSE-based method, SURF [20], optical-flow and enhanced correlation coefficient (ECC) [14]. As shown in Table II, the MSE-based method achieves better performance in terms of SCRG and BSF metrics. However, SURF and optical flow offer higher computational efficiency. Considering the trade-off between accuracy and speed, we choose MSE-based registration method in CMC.

#### IV. DISCUSSION

There are also abnormal samples generated by MSE-based registration method, in which the targets are not effectively enhanced like the correct one, the examples are shown in Fig. 6(a). Additionally, as illustrated in Fig. 6(b), the registration and frame difference steps account for over 40% of the total computational time, and the step of ME-DPC account for 51%. In contrast, the extraction of spatiotemporal contrast map is efficiency. Obviously, certain limitations remain in enhancing the real-time performance for the candidate target localization stage.

#### V. CONCLUSION

In this letter, a moving ISTD method combining the holistic interframe interaction and spatiotemporal LCM is proposed. It consists of two main steps, the first is determining the candidate target location using ME-DPC; the second is obtaining and fusing the TCM and DWESCM from the additionally temporal spatial domains to determine the location of true targets. Experiments conducted on various IR image sequences demonstrate that proposed HII-STLCM achieves competitive performance,

TABLE II

| Methods | SURF          | Optical-flow | ECC    | MSE-based     |
|---------|---------------|--------------|--------|---------------|
| SCRG    | 61.692        | 95.325       | 25.739 | <b>620.27</b> |
| BSF     | 0.4318        | 1.1017       | 0.3659 | <b>1.1229</b> |
| Time/s  | <b>0.0115</b> | 0.0248       | 0.3137 | 0.2316        |

especially for detecting infrared point-like target, in comparison to other ISTD methods. And the balance of computational efficiency and accuracy deserves further investigation, potentially with the aid of deep learning approaches [21].

#### REFERENCES

- [1] Y. Ma, Y. Liu, Z. Pan and Y. Hu, "Method of Infrared Small Moving Target Detection Based on Coarse-to-Fine Structure in Complex Scene," *Remote Sens.*, vol. 15, no. 6, p. 1508, Mar. 2023.
- [2] Y. Chen, G. Zhang, Y. Ma, J. U. Kang, and C. Kwan, "Small infrared target detection based on fast adaptive masking and scaling with iterative segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [3] D. Liu, J. Zhang, Y. Qi, Y. Xi and J. Jin, "Exploring Lightweight Structures for Tiny Object Detection in Remote Sensing Images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025.
- [4] F. Wu, H. Yu, A. Liu, J. Luo and Z. Peng, "Infrared Small Target Detection Using Spatiotemporal 4-D Tensor Train and Ring Un-folding," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–22, 2023.
- [5] C. L. P. Chen, H. Li, Y. Wei, T. Xia, and Y. Y. Tang, "A local contrast method for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 574–581, Jan. 2014.
- [6] L. Wu, Y. Ma, F. Fan, M. Wu, and J. Huang, "A double-neighborhood gradient method for infrared small target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 8, pp. 1476–1480, Aug. 2021.
- [7] C. Xia, X. Li, L. Zhao, and R. Shu, "Infrared small target detection based on multiscale local contrast measure using local energy factor," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 157–161, Jan. 2020.
- [8] J. Han et al., "A local contrast method for infrared small-target detection utilizing a tri-layer window," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1822–1826, Oct. 2020.
- [9] R. Kou et al., "Infrared Small Target Detection Based on the Improved Density Peak Global Search and Human Visual Local Contrast Mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6144–6157, 2022.
- [10] L. Z. Deng, H. Zhu, C. Tao, and Y. T. Wei, "Infrared moving point target detection based on spatial-temporal local contrast filter," *Infr. Phys. Technol.*, vol. 76, pp. 168–173, May. 2016.
- [11] P. Du and A. Hamdulla, "Infrared moving small-target detection using spatial-temporal local difference measure," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1817–1821, Oct. 2020.
- [12] Y. Zhang, K. Leng and K.-S. Park, "Infrared detection of small moving target using spatial-temporal local vector difference measure," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [13] J. Yu, L. Li, X. Li, J. Jiao, X. Su, and F. Chen, "Infrared moving small-target detection using strengthened spatial-temporal tri-layer local contrast method," *Infr. Phys. Technol.*, vol. 140, Aug. 2024.
- [14] X. Dong et al., "Detecting and Tracking of Small Infrared Targets Adaptively in Complex Background," *Aero Weaponry.*, vol. 6, pp. 6, 2019.
- [15] X. Tong, "Image registration with Fourier-based image correlation: A comprehensive review of developments and applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 4062–4081, Oct. 2019.
- [16] B. Hui et al., "A dataset for infrared image dim-small aircraft target detection and tracking underground/air background," *Sci. Data Bank*, Oct. 2019.
- [17] J.-J. Yin, H.-C. Li, Y.-B. Zheng, G. Gao, Y. Hu and R. Tao, "Spatial-temporal weighted and regularized tensor model for infrared dim and small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
- [18] X. Bai and F. Zhou, "Analysis of new top-hat transformation and the application for infrared dim small target detection," *Pattern Recognit.*, vol. 43, no. 6, pp. 2145–2156, 2010.
- [19] Y. Luo, X. Li, Y. Yan and C. Xia, "Spatial-temporal tensor representation learning with priors for infrared small target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 9598–9620, Dec. 2023.
- [20] H. Bay, T. Tuytelaars, and L. Van Gool, "SURF: Speeded up robust features," in *Proc. Eur. Conf. Comput. Vis.*, 2006, pp. 404–417.
- [21] D. Yang, H. Zhang, Y. Li, and Z. Jiang, "Label evolution based on local contrast measure for single-point supervised infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.